

ASSIGNMENT #2 DUE 9/14/22

COORDINATOR: COORDINATOR'S NAME

Section 1.6

Exercise (1.6.2 — Stefano Fochesatto). If $\varphi : G \rightarrow H$ is an isomorphism, prove that $|\varphi(x)| = |x|$ for all $x \in G$. Deduce that any two isomorphic groups have the same number of elements of order n for each $n \in \mathbb{Z}^+$. Is the result true if φ is only assumed to be a homomorphism.

Proof. Let $\varphi : G \rightarrow H$ be an isomorphism and suppose $x \in G$ such that $|x| = \alpha$. By properties of homomorphisms,

$$\varphi(x)^\alpha = \varphi(x^\alpha) = \varphi(e) = e.$$

Suppose there is a $\beta < \alpha$ such that $\varphi(x)^\beta = e$. By properties of homomorphisms it would follow that, $x^\beta = e$ which is clearly a contradiction.

Since φ is a bijection, by the previous result if there are m elements in G of order n there must be m elements in H of order n . This conclusion does not hold if φ is only a homomorphism, consider $\varphi : \mathbb{Z}_4 \rightarrow \mathbb{Z}_3$ such that $\varphi(x) = x \pmod 3$. There are two elements in $\mathbb{Z}_4$ of order 4, and none in $\mathbb{Z}_3$. □

Exercise (1.6.5 — Oleksandr Bobrovnikov). Prove that the additive groups $\mathbb{R}$ and $\mathbb{Q}$ are not isomorphic.

Solution: Suppose this two groups are isomorphic. Then there exists isomorphism $\varphi : \mathbb{R} \rightarrow \mathbb{Q}$, which is a bijection. But such bijection cannot exist as $\mathbb{R}$ is uncountable set, and $\mathbb{Q}$ is countable set. Thus this two groups are not isomorphic. □

Exercise (1.6.6 — Glen Woodworth). Prove that the additive groups $\mathbb{Z}$ and $\mathbb{Q}$ are not isomorphic.

Proof. By way of contradiction, suppose that $\mathbb{Z} \cong \mathbb{Q}$. There exists isomorphism $f : \mathbb{Q} \rightarrow \mathbb{Z}$. Then there exists $q \in \mathbb{Z}$ such that $f(q) = 1$. Observe,

$$1 = f(q) = f\left(\frac{q}{2} + \frac{q}{2}\right) = f\left(\frac{q}{2}\right) + f\left(\frac{q}{2}\right).$$

This is a contradiction because there does not exist an integer n such that $n + n = 1$. Hence $\mathbb{Z}$ and $\mathbb{Q}$ are not isomorphic. $\square$

Exercise (1.6.14 — Oleksandr Bobrovnikov). Let G and H be groups and let $\varphi : G \rightarrow H$ be a homomorphism. Define the kernel of φ to be $\{g \in G \mid \varphi(g) = 1_H\}$ (so the kernel is the set of elements in G which map to the identity of H , i.e., is the fiber over the identity of H). Prove that the kernel of φ is a subgroup (cf. Exercise 26 of Section 1) of G . Prove that φ is injective if and only if the kernel of φ is the identity subgroup of G .

Solution: To prove that $\ker \varphi$ is a subgroup in G we must show that for every $x, y \in \ker \varphi$, $xy \in \ker \varphi$, and $x^{-1} \in \ker \varphi$. As $x, y \in \ker \varphi$, $\varphi(x) = \varphi(y) = 1$. Then $\varphi(xy) = \varphi(x)\varphi(y) = 1 \cdot 1 = 1$, as φ is a homomorphism. Now as $1 = \varphi(1) = \varphi(xx^{-1}) = \varphi(x)\varphi(x^{-1}) = \varphi(x^{-1})$, we deduce that $x^{-1} \in \ker \varphi$.

Now suppose $\ker \varphi = 1$ and $\varphi(x) = \varphi(y)$. As φ is a homomorphism, $1 = \varphi(x)\varphi(y)^{-1} = \varphi(x)\varphi(y^{-1}) = \varphi(xy^{-1})$. And thus $xy^{-1} = 1$. So y^{-1} is inverse for x , and then $x = y$. So φ is injective.

Suppose φ is injective. This implies that if $\varphi(x) = \varphi(y)$, then $x = y$. Let $x \in \ker \varphi$, i.e. $\varphi(x) = 1$. We know that as φ is a homomorphism, $\varphi(1) = 1$. Then from injectivity we obtain that $x = 1$. Thus $\ker \varphi = \{1\}$. $\square$

Exercise (1.6.15 — Robert Babac). Define a map $\pi : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $\pi((x, y)) = x$. Prove that π is a homomorphism and find the kernel of π (cf. Exercise 14).

Solution: Let $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$ where $\pi((x_1, y_1)) = x_1$ and $\pi((x_2, y_2)) = x_2$. Then $\pi((x_1, y_1)(x_2, y_2)) = \pi((x_1x_2, y_1y_2)) = x_1x_2 = \pi((x_1, y_1))\pi((x_2, y_2))$. Thus, π is a homomorphism. $\ker \pi = \{(x, y) \in \mathbb{R}^2 \mid \pi((0, y)) = 0\}$ $\square$

Exercise (1.6.17 — Glen Woodworth). Let G be any group. Prove that the map from G to itself defined by $g \mapsto g^{-1}$ is a homomorphism if and only if G is abelian.

Proof. Suppose $f : G \rightarrow G$ such that $f(g) = g^{-1}$. Let $g_1, g_2 \in G$. Observe,

$$\begin{aligned} f \text{ is a homomorphism} &\iff f(g_1g_2) = f(g_1)f(g_2) \\ &\iff (g_1g_2)^{-1} = (g_1)^{-1}(g_2)^{-1} \\ &\iff (g_1g_2)(g_1g_2)^{-1} = (g_1g_2)(g_1)^{-1}(g_2)^{-1} \\ &\iff 1_G = (g_1g_2)(g_1)^{-1}(g_2)^{-1} \\ &\iff g_2g_1 = (g_1g_2)(g_1)^{-1}(g_2)^{-1}g_2g_1 \\ &\iff g_2g_1 = g_1g_2 \\ &\iff G \text{ is abelian.} \end{aligned}$$

$\square$

Exercise (1.6.18 — Robert Babac). Let G be any group. Prove that the map from G to itself defined by $g \mapsto g^2$ is a homomorphism if and only if G is abelian.

Solution: ($\Rightarrow$)

Suppose the map ϕ from G to itself is defined by $g \mapsto g^2$ is a homomorphism. Let $g_1, g_2 \in G$. Notice that $g_1g_2g_1g_2 = \phi(g_1g_2) = \phi(g_1)\phi(g_2) = g_1g_1g_2g_2$. That is, $g_1g_2g_1g_2 = g_1g_1g_2g_2$. By multiplying the inverse of g_1 to the left and the inverse of g_2 to the right to both sides of the equation, we get $g_1^{-1}g_1g_2g_1g_2g_2^{-1} = g_1^{-1}g_1g_1g_2g_2g_2^{-1} \implies g_2g_1 = g_1g_2$. Therefore, G is abelian.

($\Leftarrow$)

Suppose G is abelian. Let $g_1, g_2 \in G$. Then, $\phi(g_1g_2) = (g_1g_2)^2 = (g_1g_2)(g_1g_2) = (g_1g_1)(g_2g_2) = g_1^2g_2^2 = \phi(g_1)\phi(g_2)$. Therefore, ϕ is a homomorphism.

Thus, the map from G to itself defined by $g \mapsto g^2$ is a homomorphism if and only if G is abelian. $\square$

Exercise (1.6.19 — Glen Woodworth). Let $G = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$. Prove that for any fixed integer $k > 1$ the map from G to itself defined by $z \mapsto z^k$ is a surjective homomorphism but is not an isomorphism.

Proof. Fix $k \in \mathbb{Z}$ such that $k > 1$. Suppose $f : G \rightarrow G$ where $f(z) = z^k$. We first show that f is well-defined. Let $z_1 = z_2$ for some $z_1, z_2 \in G$. Then $f(z_1) = z_1^k = z_2^k = f(z_2)$. Thus, f is well-defined. We now show f is a surjection. Let $y \in G$. Then there exists $n \in \mathbb{Z}^+$ such that $y^n = 1$. Then $nk \in \mathbb{Z}^+$. We have $(y^{1/k})^{nk} = y^n = 1$, so that $y^{1/k} \in G$. Thus, $f(y^{1/k}) = y$, so f is surjective. We now show f is a homomorphism. Note that $G \subset \mathbb{C}$. Then, since $\mathbb{C}$ is abelian, G is abelian as well. Let $q, t \in G$. Then there exists $n_1, n_2 \in \mathbb{Z}^+$ such that $q^{n_1} = t^{n_2} = 1$. Since G is abelian, we have

$$f(qt) = (qt)^k = q^k t^k = f(q)f(t).$$

Thus, f is a homomorphism. To see that f is not an injection, note that when $z \in \mathbb{C}$, $z^k = 1$ has $k > 1$ distinct roots, all of which will be in G by definition. Hence, there are k unique elements in G such that $f(z) = z^k$, so f is not an injection. $\square$

Section 1.7

Exercise (1.7.3 — Oleksandr Bobrovnikov). Show that the additive group $\mathbb{R}$ acts on the x, y plane $\mathbb{R} \times \mathbb{R}$ by $r \cdot (x, y) = (x + ry, y)$.

Solution: We must show that for every $r, s \in \mathbb{R}$ and for every $(x, y) \in \mathbb{R}^2$ holds $s(r((x, y))) = (s + r)(x, y)$, and that $0(x, y) = (x, y)$. Indeed,

$$(1) \quad s(r(x, y)) = s(x + ry, y) = (x + ry + sy, y) = (x + (r + s)y, y) = (r + s)(x, y);$$

$$(2) \quad 0(x, y) = (x + 0y, y) = (x, y).$$

□

Exercise (1.7.4 — Stefano Fochesatto). Let G be a group acting on a set A and fix some $a \in A$. Show that the following sets are subgroups of G .

a. The kernel of the action,

Proof. Let g, h be in the kernel of the action of G and $x \in A$. Clearly the kernel of the action of G is non empty since it contains the identity. Consider gh acting on $x \in A$, since g, h are in the kernel we get the following,

$$(gh) \cdot x = (g \cdot (h \cdot x)) = (g \cdot x) = x.$$

Thus gh is in the kernel of the action of G . Therefore the group is closed with respect to function composition. Note that,

$$g^{-1} \cdot (x) = g^{-1} \cdot (g \cdot (x)) = (g^{-1}g) \cdot (x) = 1 \cdot x = x.$$

Therefore g^{-1} is in the kernel of the action of G , therefore the group is closed with respect to inverses. Thus the kernel of the action of G is a subgroup of G . □

b. $\text{Stab}_G(a) = \{g \in G \mid ga = a\}$.

Proof. Let $g, h \in \text{Stab}_G(a)$. Clearly $\text{Stab}_G(a)$ is non empty since it must contain the identity. Consider gh acting on a , since g, h are in the stabilizer group of a in G we

get,

$$(gh) \cdot a = (g \cdot (h \cdot a)) = (g \cdot a) = a.$$

Thus $gh \in \text{Stab}_G(a)$. Therefore the group is closed with respect to function composition. Note that,

$$g^{-1} \cdot (a) = g^{-1} \cdot (g \cdot (a)) = (g^{-1}g) \cdot (a) = 1 \cdot a = a.$$

Therefore $g^{-1} \in \text{Stab}_G(a)$, therefore the group is closed with respect to inverses. Thus $\text{Stab}_G(a) \subseteq G$.

□

Exercise (1.7.6 — Oleksandr Bobrovnikov). Prove that a group G acts faithfully on a set A if and only if the kernel of the action is the set consisting only of the identity.

Solution: An action is faithful if the only element $g \in G$ such that for every $a \in A$, $ga = a$ is identity element. Kernel is the set of the elements of G which fix all the elements of A . So action is faithful iff kernel consists only of the identity element. □

Exercise (1.7.15 — Stefano Fochesatto). Let G be any group and let $A = G$. Show that the maps defined by $g \cdot a = ag^{-1}$ for all $g, a \in G$ do satisfy the axioms of a left group action of G on itself.

Proof. Let G be any group and let $A = G$. Suppose the maps defined by $g \cdot a = ag^{-1}$ for all $g, a \in G$. Consider,

$$(gh) \cdot a = a(gh)^{-1} = a(h)^{-1}(g)^{-1} = (g \cdot (h \cdot a)).$$

Also note,

$$1 \cdot a = a(1)^{-1} = a$$

Therefore $g \cdot a = ag^{-1}$ defines a group action on itself. $\square$

Exercise (1.7.16 — Robert Babac). Let G be a group and let $A = G$. Show that the map defined by $g \cdot a = gag^{-1}$ for all $g, a \in G$ do satisfy the axioms of a (left) group action (this action of G on itself is called conjugation).

Solution: Let $g_1, g_2, a, e \in G$ where e is the identity element of G . Then, $g_1 \cdot (g_2 \cdot a) = g_1 \cdot g_2 a g_2^{-1} = g_1 g_2 a g_2^{-1} g_1^{-1} = (g_1 g_2) a (g_1 g_2)^{-1} = (g_1 g_2) \cdot a$, and $e \cdot a = e a e^{-1} = a$. Therefore, the map defined by $g \cdot a = gag^{-1}$ for all $g, a \in G$ do satisfy the axioms of a (left) group action. $\square$

Exercise (1.7.17 — Noah Palmer). Let G be a group and let G act on itself by left conjugation, so each $g \in G$ maps G to G by

$$x \mapsto gxg^{-1}$$

For fixed $g \in G$, prove that conjugation by g is an isomorphism from G onto itself (i.e, is an automorphism of G). Deduce that x and gxg^{-1} have the same order for all $x \in G$ and that for any subset A of G , $|A| = |gAg^{-1}|$ (here $gAg^{-1} = \{gag^{-1} \mid a \in A\}$).

Proof. Let G be a group. For some fixed $g \in G$ we prove that the left conjugation by g , that is the function given by $\phi_g(a) = gag^{-1}$, is an isomorphism from G to itself. Let $a, b \in G$ and consider that

$$\phi_g(ab) = gabg^{-1} = ga(g^{-1}g)bg^{-1} = \phi_g(a)\phi_g(b).$$

Hence, ϕ_g is a homomorphism. We now prove it is a bijection by showing that it is both injective and surjective. To show it is injective we show that $\ker(\phi) = \{1_G\}$, assume for the sake of contradiction there exists $x \in G$ such that $\phi_g(x) = 1_G$ and $x \neq 1_G$. This implies that $gxg^{-1} = 1_G$. Thus, $gx = g$ and so $x = 1_G$, a contradiction. Therefore, the kernel of ϕ_g is the set $\{1_G\}$. So, ϕ_g is an injection. Next to show that ϕ_g is surjective, choose $y \in G$. Then $g(g^{-1}yg)g^{-1} = y$, hence ϕ_g is surjective, and thus a bijection. Because ϕ_g is a bijection and

a homomorphism it follows that it is an isomorphism.

To show that x and gxg^{-1} have the same order, assume that $|x| = n$. Then $\phi(x^n) = gx^n g^{-1} = gxg^{-1}g \dots gxg^{-1} = (gxg^{-1})^n = 1_G$. Thus, $|gxg^{-1}| \leq n$. To show that $|gxg^{-1}| = n$, suppose there exists $m < n$ such that $(gxg^{-1})^m = 1_G$. This would imply that $\phi(x^m) = 1_G$ which is a contradiction since $|x| = n > m$. Hence, $|x| = |gxg^{-1}|$ for all $x \in G$ and so $|A| = |gAg^{-1}|$. $\square$

Exercise (1.7.18 — Noah Palmer). Let H be a group acting on a set A . Prove that the relation $\sim$ defined by

$$a \sim b \text{ if and only if } a = hb \text{ for some } h \in H$$

is an equivalence relation. (For each $x \in A$ the equivalence class of x under $\sim$ is called the orbit of x under the action of H . The orbits under the action of H partition the set A).

Proof. Let H be a group acting on some set A , we prove that the relation $\sim$ defined by $a \sim b$ if and only if $a = hb$ for some $h \in H$ is an equivalence relation. First, observe that $a \sim a$ since $a = 1_H a$, and so $\sim$ is reflexive. Next, let $a \sim b$, that is $a = hb$ for some $h \in H$. It follows that $h^{-1}a = b$ and so $b \sim a$. Therefore, $\sim$ is symmetric. Finally, let $a \sim b$ and $b \sim c$. Then $a = hb$ and $b = h'c$ for some $h, h' \in H$. Thus, $a = hh'c$ which implies that $a \sim c$. Hence, $\sim$ is transitive. Since $\sim$ is reflexive, symmetric and transitive we conclude that $\sim$ is by definition an equivalence relation. Therefore, the orbits, that is the equivalence class of x under $\sim$, under the action of H partition the set A . $\square$

Exercise (1.7.19 — Noah Palmer). Let H be a subgroup of the finite group G and let H act on G by left multiplication. Let $x \in G$ and let $\mathcal{O}$ be the orbit of x under the action of H . Prove that the map

$$H \rightarrow \mathcal{O} \text{ defined by } h \mapsto hx$$

is a bijection (hence all orbits have cardinality $|H|$). From this and the preceding exercise deduce Lagrange's Theorem:

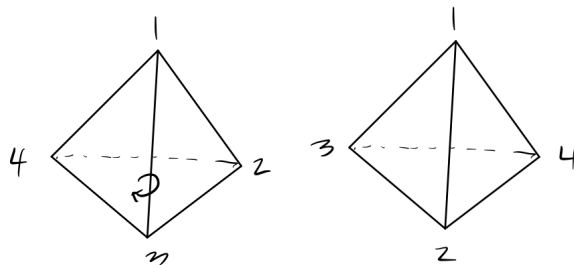
if G is a finite group and H is a subgroup of G then $|H|$ divides $|G|$.

Proof. Let H be a subgroup of some finite group G and let H act on G by left multiplication. Let $x \in G$ we prove that the map $\phi : H \rightarrow \mathcal{O}$ defined by $h \mapsto hx$ is a bijection. To show that it is surjective choose $y \in \mathcal{O}$. By definition $y = gx$ for some $g \in H$ and so $\phi(g) = gx = y$. Thus, ϕ is surjective. Next, suppose $\phi(a) = \phi(b)$. It follows that $ax = bx$ and so $axx^{-1} = bxx^{-1}$ which implies that $a = b$. Thus, ϕ is injective. Since ϕ is both injective and surjective it follows that ϕ is a bijection. Thus, for all $\mathcal{O} \in G$ $|\mathcal{O}| = |H|$.

By the preceding exercise the group action given by ϕ partitions G into orbits $\mathcal{O}_1, \dots, \mathcal{O}_n$ for $n \in \mathbb{N}$, and so $|G| = |\mathcal{O}_1| + \dots + |\mathcal{O}_n|$. But since we just proved that for all k , $|\mathcal{O}_k| = |H|$ it follows that $|G| = n|H|$. Therefore, the order of any subgroup H of G divides $|G|$. $\square$

Exercise (1.7.20 — Robert Babac). Show that the group of rigid motions of a tetrahedron is isomorphic to a subgroup (cf. Exercise 26 of Section 1) of S_4 .

Solution: Let us first find all the possible rigid motions of a tetrahedron that maps to a unique permutation of 4 elements, let's say $\{1, 2, 3, 4\}$. Notice that 1 rigid motion of the tetrahedron consists of fixing 1 vertex and "rotating" the other three vertices (i.e., if we fix vertex a , then for vertex b, c, d , $b \rightarrow c$, $c \rightarrow d$, and $d \rightarrow a$). For example,



We can get the trivial rigid motions mapping to the permutations

$$(), (234), (432), (134), (431), (124), (143)(123), \text{ and } (321).$$

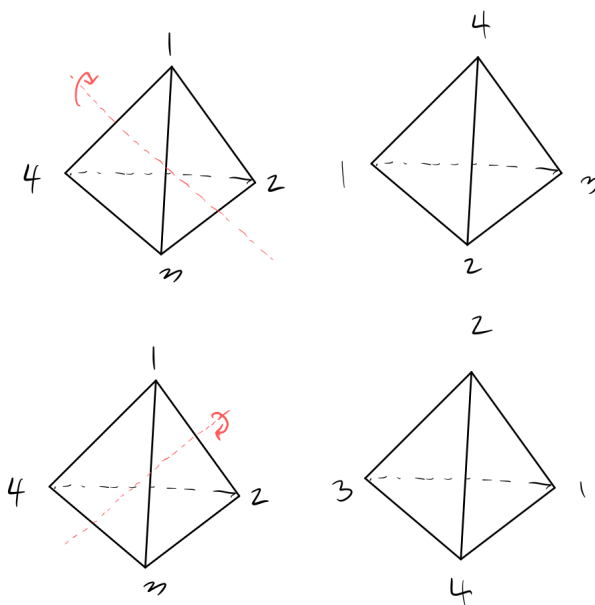
Note, $()$ is the permutation corresponding to the rigid motion where nothing happens. Notice that for a rigid motion, if we fix vertex a, we cannot also fix vertex b and have only vertices c and d swap places. Thus, we cannot have a rigid motion mapping to the permutations with 1 transposition. That is, we cannot have

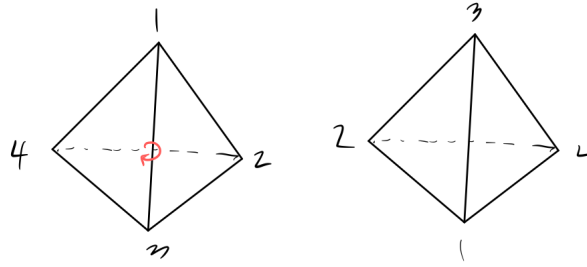
$$(12), (13), (14), (23), (24), \text{ and } (34).$$

By further inspection, we can say any permutation of length 4 will also not be an allowed rigid motion since the decomposition of the permutation consists of the permutation of a rigid motion with 1 transposition. For example, $(1234) = (123)(14)$. Thus, the permutations

$$(1234), (1243), (1324), (1342), (1423), \text{ and } (1432)$$

are also not considered rigid motions. We can find the last three rigid motions by below.





Thus, the last three rigid motions correspond to the permutations,

$$(12)(34), (13)(24), \text{ and } (14)(23).$$

Thus, the group of rigid motions of a tetrahedron is isomorphic to the group of even permutations of S_4 . $\square$

Section 2.1

Exercise (2.1.5 — Glen Woodworth). Prove that G cannot have a subgroup H with $|H| = n - 1$ where $n = |G| > 2$.

Proof. Suppose $n = |G| > 2$ and $H \subseteq G$ where $|H| = n - 1$. Then $n - 1 > 1$. We have

$$n - 1 \equiv 0 \pmod{n - 1}.$$

Adding 1 to both sides, we have

$$n \equiv 1 \pmod{n - 1}.$$

Hence, $|H| = n - 1$ does not divide $|G| = n$, so by Lagrange's Theorem, H is not a subgroup of G . Since H was chosen arbitrarily from the possible subsets of G with cardinality $n - 1$, any subset of G with a cardinality of $n - 1$ is not a subgroup of G . $\square$

Exercise (2.1.6 — Oleksandr Bobrovnikov). Let G be an abelian group. Prove that $\{g \in G \mid |g| < \infty\}$ is a subgroup of G (called the torsion subgroup of G). Give an explicit example where this set is not a subgroup when G is non-abelian.

Solution: Let $H = \{g \in G \mid |g| < \infty\}$. We must show that for every $h_1, h_2 \in H$, $h_1 h_2 \in H$. Let $h_1, h_2 \in H$. Then there exist $m, n \in \mathbb{Z}$ such that $h_1^m = h_2^n = 1$. Then, as h_1 and h_2 commute, $(h_1 h_2)^{mn} = h_1^{mn} h_2^{mn} = 1^n \cdot 1^m = 1$. Inverse element for h_1 is h_1^{m-1} . Thus torsion subgroup is closed under multiplication and inverses, and is indeed a subgroup of G .

Consider a group $D_{2,\infty} = \langle r, s \mid s^2 = 1, sr = s^{-1}s \rangle$. Then $|s| = 2$ and $|sr| = 2$ as $(sr)^2 = sr sr = r^{-1} s s r = 1$, But $s \cdot sr = r$, and $|r| = +\infty$. Thus torsion set contains elements s and sr , but is not closed under multiplication, as their product has infinite order. So it is not a subgroup. $\square$

Exercise (2.1.8 — Stefano Fochesatto). Let H and K be subgroups of G . Prove that $H \cup K$ is a subgroup if and only if either $H \subseteq K$ or $K \subseteq H$.

Proof. Let H and K be subgroups of G . Suppose that $H \cup K$ is a subgroup and without loss of generality let $H \not\subseteq K$. Therefore there exists some $h \in H$ such that $h \notin K$. Consider $k \in K$ and note that $hk \in H \cup K$ and since $h \notin K$, $hk \notin K$ and therefore it follows that $hk \in H$. Since H is a subgroup $(h^{-1}h)k = k \in H$ and therefore $K \subseteq H$.

For the reverse statement suppose without loss of generality that $K \subseteq H$. Therefore $H \cup K = H$, which is by definition a subgroup. $\square$

Section 2.2

Exercise (2.2.6 — Glen Woodworth). Let H be a subgroup of the group G .

- (1) Show that $H \leq N_G(H)$. Give an example to show that this is not necessarily true if H is not a subgroup.
- (2) Show that $H \leq C_G(H)$ if and only if H is abelian.

Proof. (a) Since $H \leq G$, $H \neq \emptyset$. Suppose $h_1 \in H$. For all $h \in H$, we have $hh_1h^{-1} \in H$ because H must be closed under the binary operation of G . Hence, $H \subseteq N_G(H)$, but since $H \leq G$, $H \leq N_G(H)$. As an example, suppose $G = S_4$ and $H = \{1, (12)\}$. Then (12) has

no inverse in H , so H is not a subgroup of $C_G(H)$.

(b) First, note that since $H \leq G$, and $C_G(H) \leq G$, we have

$$H \subseteq C_G(H) \iff H \leq C_G(H).$$

Now, observe

$$\begin{aligned} H \leq C_G(H) &\iff H \subseteq C_G(H) \\ &\iff \forall h_1, h_2 \in H, h_2 h_1 h_2^{-1} = h_1 \\ &\iff \forall h_1, h_2 \in H, h_2 h_1 = h_1 h_2 \\ &\iff H \text{ is abelian.} \end{aligned}$$

□

Exercise (2.2.7 — Noah Palmer). Let $n \in \mathbb{Z}$ with $n \geq 3$. Prove the following:

- (a) $Z(D_{2n}) = 1$ if n is odd
- (b) $Z(D_{2n}) = \{1, r^k\}$ if $n = 2k$

Proof. (a) Let $n \in \mathbb{Z}$ where n is a odd and $n \geq 3$. Assume for the sake of contradiction that there exists $g \in Z(D_{2n})$ and $g \neq 1$. Thus, $gx = xg$ for all $x \in G$. We know that $g = s^i r^k$ for some $0 \leq k < n$ and $i = 0$ or 1 . It follows that $s^i r^k s = s^{i+1} r^k$ which implies that $s^{i+1} r^{-k} = s^{i+1} r^k$ and so $r^k = r^{-k}$. Thus, $r^{2k} = 1$. But this is a contradiction since n is odd and we assumed that $k < n$. Hence, there exists no such elements, and so $Z(D_{2n}) = 1$.

(b) Let $n = 2k$ for some $k \in \mathbb{Z}$. Observe that $r^k r^k = 1$, so $r^k = r^{-k}$. Now, choose $a \in D_{2n}$ and so a can be expressed in the form $a = s^i r^j$ for $j \in \mathbb{Z}$ and $i = 0$ or 1 . Then $r^k s^i r^j r^k = s^i r^j r^{k-k} = s^i r^j$, and so $r^k \in Z(D_{2n})$. To show that r^k and 1 are the only elements in $Z(D_{2n})$, assume for the sake of contradiction that there exists $b = s^i r^m$ such that $bx = xb$ for all $x \in D_{2n}$. Thus, $s^{i+1} r^m = s^i r^m s$ for $0 \leq m < 2k$ and $i = 0$ or 1 . If $i = 0$ then $r^m s = s r^{-m} = s r^m$ which implies that $r^m = r^{-m}$ implying

that $m = k$, and so $b = r^k$. Instead, if $i = 1$ then $sr^m s = s^2 r^{-m} = r^{-m} = r^m = s^2 r^m$, and so again $b = r^k$. In either, case b is equal to r^k which is a contradiction since we assumed that b was a distinct element of $Z(D_{2n})$. Thus, no such element exists and $Z(D_{2n}) = \{1, r^k\}$.

□

Exercise (2.2.9 — Noah Palmer). For any subgroup H of G and any nonempty subset A of G define $N_H(A)$ to be the set $\{h \in H \mid hAh^{-1} = A\}$. Show that $N_H(A) = N_G(A) \cap H$ and deduce that $N_H(A)$ is a subgroup of H (note that A need not be a subset of H).

Proof. Let H be some subgroup of G and let A be some nonempty subset of A . Choose $a \in N_H(A)$. It follows by definition that $a \in H$ and $a \in G$. Since $a \in G$ and $aAa^{-1} = A$ it follows that $a \in N_G(A)$. Therefore, $a \in N_G(A) \cap H$. Next choose $b \in N_G(A) \cap H$. It follows that $b \in H$ and $bAb^{-1} = A$ and so by definition $b \in N_H(A)$. Thus, $N_H(A) = N_G(A) \cap H$. Since for all $h \in N_H(A)$, $h \in H$ it follows that $N_H(A) \leq H$. □

Exercise (2.2.10 — Stefano Fochesatto). Let H be a subgroup of order 2 in G . Show that $N_G(H) = C_G(H)$. Deduce that if $N_G(H) = G$ then $H \subseteq Z(G)$.

Proof. Let H be a subgroup of G and suppose $H = \{1, h\}$. Consider $g \in N_G(H)$, by definition $gHg^{-1} = H$. By properties of group actions we know that $g1g^{-1} = 1$ and since there is only one other element in H we know that $ghg^{-1} = h$. Therefore $g \in C_G(H)$ and $N_G(H) \subseteq C_G(H)$. Consider $g \in C_G(H)$. By definition for all $a \in H$, $gag^{-1} = a$ and therefore $gHg^{-1} = H$. It follows that $g \in N_G(H)$, therefore $C_G(H) \subseteq N_G(H)$. Thus $N_G(H) = C_G(H)$. If $N_G(H) = G$ by what we have shown $C_G(H) = G$. Therefore for all $h \in H$ and $g \in G$, $ghg^{-1} = h$ which by right cancellation implies $gh = hg$. Since H is a subgroup of G , $h \in G$ and therefore by definition $H \subseteq Z(G)$. □

Exercise (2.2.11 — Robert Babac). Prove that $Z(G) \leq N_G(A)$ for any subset A of G .

Solution: Let G be a group and A be a subset of G . Let $z_1, z_2 \in Z(G)$. Then, $z_1g = gz_1$ and $z_2g = gz_2$ for all $g \in G$. Observe that $(z_1z_2)g = z_1(z_2g) = z_1(gz_2) = (z_1g)z_2 = (gz_1)z_2 = g(z_1z_2)$. That is, $Z(G)$ is closed. Notice that $z_1g = gz_1 \implies z_1^{-1}z_1gz_1^{-1} = z_1^{-1}gz_1z_1^{-1} \implies gz_1^{-1} = z_1^{-1}g$ for all $g \in G$. Thus, inverses are in $Z(G)$. Therefore, $Z(G) \leq G$. $\square$